Machine Learning Engineering Bootcamp

Capstone: Deployment Architecture

Step 10

Learning Objective

- Create a concrete plan for taking the scaled prototype from Phase 1 into production.

Criteria	Meets Expectations	Weightage
Time Estimate	4 - 8 hours	
Completion	❏ 1 - 2 page Google document (commentable by mentor). ❏ An architecture diagram (as a drawing or sketch).	1 point
Process and understanding	❏ The submission demonstrates that the student has articulated a deployment architecture that answers the following questions about the deployment of the Capstone Project prototype (as relevant): ● What are the major components of your system? What are the inputs and outputs? ● Where and how will the data be stored?	7 points

	- How will data get from one component of the system to another? - What is the lifecycle of the ML/DL model? - How frequently does the model need retraining? Is it at fixed intervals or when certain conditions are met? - What kind of data is needed for retraining? How will the data be stored and managed? - How will the retrained model be evaluated? - How will the retrained model be deployed? - How will the retrained model be stored as an artifact? - How will the system be monitored and debugged? - What are the specific tools/technologies that will be used to build this system? - What is the estimated implementation cost in terms of	

	resources, time, and money as applicable?	
Presentation	❏ The architecture diagram is clear and labeled. ❏ The document clearly explains the major decisions that were made with justification.	2 points

Excellence: The above criteria were met and:

- *The submission describes how the architecture can handle various edge and stress cases, and how it can be changed to accommodate those.*
- *The student has put a lot of thought into how to make the application scalable, low-latency, fault-tolerant, and resilient.*
- *The student has proposed multiple viable architectures based on different usage scenarios or resource constraints*